IECEE OD-5002-3

Edition 1.0 2015-06-03

IECEE
OPERATIONAL DOCUMENT

IEC System of Conformity Assessment Schemes for Electrotechnical Equipment and Components (IECEE System)

Committee of Testing Laboratories (CTL)

CTL Provisional Decision Sheet (PDSH)



IECEE OD-5002-3:2015(EN)



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INTERNATIONAL
ELECTROTECHNICAL
COMMISSION

PRICE CODE ZZ

CTL PROVISIONAL DECISION SHEET (PDSH)

Standard(s) (incl. year)	Subclause(s)	Tracking No.	Year
IEC 60950-1: 2005 + Am1:2009 + Am2:2013 IEC 60065:2001 + Am1:2005 + Am2:2010 IEC 60065:2014	2.1.1.7 9.1.6	PDSH 1080A	2017
Category			
OFF, TRON			
Subject	**Keywords**	**Developed by**	**To be approved**
Zero loss automatic X cap discharge	IC including capacitor discharge function (ICX) IC, capacitor discharge, Zero loss	ETF2	(Year) CTL Plenary Meeting

Question

How shall we test equipment with IC including a capacitor discharge function?

Decision

Since requirements for ICX are not in IEC 60065 and IEC 60950-1, use of such an IC is a “construction not specifically covered,” such as described in 1.3.4 of IEC 60950-1. Since G.16 of IEC 62368-1 is the current thinking of IEC TC108 and demonstrates “a level of safety not less than that generally afforded by this standard and the principles of safety contained herein, “ it also is appropriate to use G.16 for such constructions in IEC 60065 and IEC 60950-1.

~~IC including capacitor discharge function (ICX)~~

~~Requirements~~

~~An ICX and any associated components critical to the discharge function of a capacitor to an accessible part (such as the mains capacitor) are not fault tested if one of the following conditions is met:~~

- ~~the ICX with the associated circuitry as provided in the equipment complies with the tests below. Any impulse attenuating components (such as varistors and GDT’s) that attenuate the impulse to the ICX and the associated circuitry are disconnected.~~
  ~~If discharge components external to the ICX are necessary, they shall not fail during the tests; or~~
- ~~the ICX tested separately complies with the requirements and tests below. If discharge components external to the ICX are necessary:~~
  - ~~they shall be included in the test; and~~
  - ~~they shall not fail during the tests; and~~
  - ~~the discharge components used in the equipment shall be within the range tested~~

~~Tests~~

Where the ICX is tested by itself, the test set up shall be as recommended by the ICX manufacturer.

- humidity treatment for 120 h at a temperature of (40±2)°C and a relative humidity of (93±3)%
- 100 positive impulses and 100 negative impulses between line and neutral using a capacitor with the largest capacitance and a resistor with the smallest resistance specified by the manufacturer of the ICX; and repeated with a capacitor with the smallest capacitance and the resistor with the largest resistance. The time between any two impulses shall not be less than 1 s. The impulse shall be as specified in circuit 2 of Table N.1 (60950-1) / 1,2/50 µs in Table K.1 (60065), with Uc equal to the transient voltage.
- Application of an a.c. voltage that is 110 % of the rated voltage for 2,5 minutes.
- 10 000 cycles of power on and off using a capacitor with the smallest capacitance and a resistor with the largest resistance as specified by the manufacturer of ICX. The power on and off cycles time shall not be less than 1 s.
- If any of the associated circuitry components other than those critical for the discharge function fails, it may be replaced with a new component

Compliance criteria

Compliance is checked by evaluation of the available data or by conducting the above tests. The capacitor discharge test is conducted after the above tests, ensuring the ICX or the EUT provided with the ICX continues to provide the **safeguard** function.

NOTE Evaluation of available data should include information of failure of any associated circuitry components keeps the discharge mode in the on/stay mode

Explanatory notes

IC chips handling discharge of capacitors in equipment to cope with environmental issues: (Limited power consumption in standby conditions)

Product Highlights for such IC's

- Blocks current through X-cap discharge resistor when AC voltage is connected
- Automatically discharges X-cap through discharge resistors when AC is disconnected

Single fault condition (open circuit conditions) of the IC and associated circuitry components critical for the discharge function is the concern related to safety, when AC power is disconnected. If the capacitance of X-capacitors connected in the equipment is above a certain value, it would result in a noncompliance with the requirements for plug discharge in IEC 60950-1 2.1.1.7 / IEC 60065 9.1.6.

IEC TC108 HBSDT agreed to include requirements for IC including capacitor discharge function (ICX) to the FDIS for IEC 62368-1 2ed. The proposal below is in accordance with this.

Amd.2 to IEC 60950-1 ed2 and IEC 60065 ed8 (assumed to be the last updates to these std.) do not include these requirements. IEC TC108 MT1/MT2 has therefore been asked to confirm this CTL proposal.

Recommendation by IEC TC108 MT1/MT2, 10 May 2013:
The TC108 management discussed the issue and feels that the approach to use the requirements from the latest draft IEC 62368-1, edition 2 is very reasonable. It is also in line with the statements in the drafts of IEC 60065 and IEC 60950-1 that allow components and subassemblies to be used that are evaluated according to IEC 62368-1.

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